

List

- a list is an ordered and stored multiple values .
- its is mutable colletion of the item or changable.
- list can store elements of the different datatype. (int, float, string, object)
- allow dulplicate values.

adding methods

1. append() :- Adds an elements to the end of the list.
2. insert() :- insert an elements at the specific index.
3. .extented :- add all elements of the an iterable to the end.

removal method

4. .remove() :-remove the first occurence of the values
5. .pop() :- remove and return the elements at a given index(default last).
6. .clear() :- remove all elements .

searching and checking method

7. .index():- Return the index of first machting elements .
8. .count():- Counts occurrence of the an elements .

sorting and reversing method

9. sorting() :- Sorting a sorts the list place (deafault as ascending).
10. .reverse():- descending the item.

coping method

11. .copy():-create a shallow copy.

```
In [2]: a=[] # creating a list
a.append(1) # multiple values will not accepted
```

```
a.append(2)
a.append(3)
print(a)
```

[1, 2, 3]

```
In [3]: a=[1,3]
a.insert(1,2) # at index 1 add 2 value
print(a)
```

[1, 2, 3]

```
In [4]: a=[1,2]
a.extend( [3,4,5])
print(a)
```

[1, 2, 3, 4, 5]

```
In [5]: a=[1,2,2,3]
a.remove(2)
print(a)
```

[1, 2, 3]

```
In [27]: l=[10,20,30] #bydefault last element
x= l.pop()
print(x)
print(l)
```

30

[10, 20]

```
In [29]: l=[10,20,30]
x= l.pop(1)
print(x)
print(l)
```

20

[10, 30]

```
In [18]: a=[1,2,3]
a.clear()
print(a)
```

[]

```
In [23]: a=["a", 'b', 'c', "b"]
print(a.index("b"))
```

1

1

```
In [30]: a= [1,2,3,4,4,4,5,5,6,2,1,1,1,2,3]
print(a.count(2))
print(a.count(4))
print(a.count(5))
```

3

3

2

```
In [1]: a=[1,8,4,5,6,8,43,47,54]
a.sort()
```

```
print(a)
```

```
[1, 4, 5, 6, 8, 8, 43, 47, 54]
```

```
In [5]: a=[1,8,4,5,6,8,43,47,54]
a.sort(reverse=True)
print(a)
```

```
[54, 47, 43, 8, 8, 6, 5, 4, 1]
```

```
In [6]: a=[1,2,3]
a.reverse()
print(a)
```

```
[3, 2, 1]
```

```
In [7]: a=[1,2,3]
b=a.copy()
print(a)
print(b)
```

```
[1, 2, 3]
```

```
[1, 2, 3]
```

```
In [8]: a==b
```

```
Out[8]: True
```

```
In [15]: # nested indexing

l=[1,2,3,"nit",True]
print(l.index('nit'))
print(l[3][0])
print(l[3][1])
print(l[3][2])
```

```
3
```

```
n
```

```
i
```

```
t
```

```
In [16]: # practise
```

```
In [49]: #      creating a list
l=[ ]
numbers=[1,2,3,4]
mixed= [1,'hello',3.14,True]
```

```
In [53]: print(mixed*3)
```

```
[1, 'hello', 3.14, True, 1, 'hello', 3.14, True, 1, 'hello', 3.14, True]
```

```
In [51]: # accessing elements
numbers[0]
```

```
Out[51]: 1
```

```
In [45]: numbers[-1]
```

```
Out[45]: 4
```

```
In [46]: # modifying a list
numbers.append(5)
numbers.insert(0,0)
numbers[4]= 100
print(numbers)
```

```
[0, 1, 2, 3, 100, 5]
```

```
In [52]: #renoving elements
numbers.remove(3)
numbers.remove(2)
numbers.clear()
print(numbers)
```

```
[]
```

```
In [55]: # useful list operations
numbers=[1,2,3,4]
mixed= [1,'hello',3.14,True]

print(len(mixed))
print(sum(numbers))
print(sorted(numbers))
print(numbers.reverse())
del numbers
```

```
4
```

```
10
```

```
[1, 2, 3, 4]
```

```
None
```

```
In [59]: #Looping through a list
for i in mixed:
    print(i)
```

```
1
```

```
hello
```

```
3.14
```

```
True
```

```
In [ ]:
```